

There are five constraints (different from the non negativity constraints), so that there are five dual variables. As all constraints are $\leq$, the corresponding dual variables are non negative, according to the summary table.

$$\begin{array}{ll}
-5x_1 + 4x_2 \leq 16, & \gamma_1 \leq 0, \\
x_1 + 3x_2 \leq 31, & \gamma_2 \leq 0, \\
x_1 + 2x_2 \leq 24, & \gamma_3 \leq 0, \\
3x_1 + 5x_2 \leq 68, & \gamma_4 \leq 0, \\
x_1 + x_2 \leq 122, & \gamma_5 \leq 0.
\end{array}$$

As the two variables of the primal are non negative, the constraints of the dual are $\leq$, according to the summary table. Therefore, the dual problem is:

$$\max 16\gamma_1 + 31\gamma_2 + 24\gamma_3 + 68\gamma_4 + 122\gamma_5 \quad (1)$$

subject to

$$\begin{array}{ll}
-5\gamma_1 + \gamma_2 + \gamma_3 + 3\gamma_4 + \gamma_5 \leq -4, \\
4\gamma_1 + 3\gamma_2 + 2\gamma_3 + 5\gamma_4 + \gamma_5 \leq -7, \\
\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5 \leq 0.
\end{array} \quad (2)$$

In order to calculate the Lagrangian, we write the problem as

$$\min -4x_1 - 7x_2$$

subject to

$$\begin{array}{ll}
g_1(x) = -16 - 5x_1 + 4x_2 \leq 0 & (\mu_1) \\
g_2(x) = -31 + x_1 + 3x_2 \leq 0 & (\mu_2) \\
g_3(x) = -24 + x_1 + 2x_2 \leq 0 & (\mu_3) \\
g_4(x) = -68 + 3x_1 + 5x_2 \leq 0 & (\mu_4) \\
g_5(x) = -122 + x_1 + x_2 \leq 0 & (\mu_5) \\
g_6(x) = -x_1 \leq 0 & (\mu_6) \\
g_7(x) = -x_2 \leq 0 & (\mu_7).
\end{array}$$

The Lagrangian function of this problem is

$$\begin{aligned}
L(x_1, x_2, \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7) &= -4x_1 - 7x_2 + \mu_1(-16 - 5x_1 + 4x_2) \\
&\quad + \mu_2(-31 + x_1 + 3x_2) + \mu_3(-24 + x_1 + 2x_2) \\
&\quad + \mu_4(-68 + 3x_1 + 5x_2) + \mu_5(-122 + x_1 + x_2) \\
&\quad - \mu_6x_1 - \mu_7x_2 \\
&= x_1(-4 - 5\mu_1 + \mu_2 + \mu_3 + 3\mu_4 + \mu_5 - \mu_6) \\
&\quad + x_2(-7 + 4\mu_1 + 3\mu_2 + 2\mu_3 + 5\mu_4 + \mu_5 - \mu_7) \\
&\quad - 16\mu_1 - 31\mu_2 - 24\mu_3 - 68\mu_4 - 122\mu_5.
\end{aligned}$$

In order for the dual function to be bounded, the coefficients of x_1 and x_2 have to be zero:

$$\mu_6 = -4 - 5\mu_1 + \mu_2 + \mu_3 + 3\mu_4 + \mu_5, \quad (3)$$

$$\mu_7 = -7 + 4\mu_1 + 3\mu_2 + 2\mu_3 + 5\mu_4 + \mu_5, \quad (4)$$

and the dual function is

$$q(\mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7) = -16\mu_1 - 31\mu_2 - 24\mu_3 - 68\mu_4 - 122\mu_5.$$

Moreover, we must impose $\mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7 \geq 0$. Thus, As μ_6 and μ_7 do not appear in the dual function, they can be eliminated.

- Combining (3) and $\mu_6 \geq 0$, we obtain $-4 - 5\mu_1 + \mu_2 + \mu_3 + 3\mu_4 + \mu_5 \geq 0$.
- Combining (4) and $\mu_7 \geq 0$, we obtain $-7 + 4\mu_1 + 3\mu_2 + 2\mu_3 + 5\mu_4 + \mu_5 \geq 0$.

Therefore, the constraints on the dual variables are

$$\begin{aligned} -5\mu_1 + \mu_2 + \mu_3 + 3\mu_4 + \mu_5 &\geq 4, \\ 4\mu_1 + 3\mu_2 + 2\mu_3 + 5\mu_4 + \mu_5 &\geq 7, \\ \mu_1 &\geq 0, \\ \mu_2 &\geq 0, \\ \mu_3 &\geq 0, \\ \mu_4 &\geq 0, \\ \mu_5 &\geq 0. \end{aligned}$$

The dual problem is written as

$$\max_{\mu \in \mathbb{R}^5} -16\mu_1 - 31\mu_2 - 24\mu_3 - 68\mu_4 - 122\mu_5$$

subject to

$$\begin{aligned} -5\mu_1 + \mu_2 + \mu_3 + 3\mu_4 + \mu_5 &\geq 4, \\ 4\mu_1 + 3\mu_2 + 2\mu_3 + 5\mu_4 + \mu_5 &\geq 7, \\ \mu_1 &\geq 0, \\ \mu_2 &\geq 0, \\ \mu_3 &\geq 0, \\ \mu_4 &\geq 0, \\ \mu_5 &\geq 0. \end{aligned}$$

If we define $\gamma_i = -\mu_i$, $i = 1, \dots, 5$, we obtain

$$\max_{\mu \in \mathbb{R}^5} 16\gamma_1 + 31\gamma_2 + 24\gamma_3 + 68\gamma_4 + 122\gamma_5$$

subject to

$$\begin{aligned} 5\gamma_1 - \gamma_2 - \gamma_3 - 3\gamma_4 - \gamma_5 &\geq 4, \\ -4\gamma_1 - 3\gamma_2 - 2\gamma_3 - 5\gamma_4 - \gamma_5 &\geq 7, \\ -\gamma_1 &\geq 0, \\ -\gamma_2 &\geq 0, \\ -\gamma_3 &\geq 0, \\ -\gamma_4 &\geq 0, \\ -\gamma_5 &\geq 0, \end{aligned}$$

which corresponds to the optimization problem (1)–(2) obtained using the summary tables.